Model selection with nested sampling

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Model selection

Model selection with nested sampling
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Tutorial
- Github repository
- License
- Statistics

Data
- HBV.nex

XML
- HBVStrict-NS.xml
- HBVStrict-NS30.xml
- HBVStrict.xml
- HBVUCLN-NS.xml
- HBVUCLN-NS30.xml
- HBVUCLN.xml

Output
- HBVStrict-NS30.log
- HBVStrict-NS30.out
- HBVStrict.log
- HBVUCLN-NS30.log
- HBVUCLN-NS30.out
- HBVUCLN.log

Background

BEAST provides a bewildering number of models. Bayesians have two techniques to deal with this problem: model averaging and model selection. With **model averaging**, the MCMC algorithm jumps between different models, and models more appropriate for the data will be sampled more often than unsuitable models (see for example the substitution model averaging tutorial). **Model selection** on the other hand just picks one model and uses only that model during the MCMC run. This tutorial gives some guidelines on how to select the model that is most appropriate for your analysis.

Bayesian model selection is based on estimating the marginal likelihood: the term forming the denominator in Bayes' formula. This is generally a computationally intensive task and there are several ways to estimate them. Here, we concentrate on nested sampling as a way to estimate the marginal likelihood as well as the uncertainty in that estimate.

Say, we have two models, M_1 and M_2, and estimates of their (log) marginal likelihoods, Z_1 and Z_2. We can calculate the Bayes' factor (BF), which is the fraction $BF = Z_1/Z_2$ (or the difference $\log(BF) = \log(Z_1) - \log(Z_2)$ in log space). Typically, if $\log(BF)$ is greater than 1, M_1 is favoured, otherwise M_1 and $M2$ are hardly distinguishable or M_2 is favoured. How much M_1 is favoured over M_2 can be found in the following table (Kass & Raftery, 1995):

BF range	$\ln(BF)$ range	$\log_{10}(BF)$ range	Interpretation

Learning objectives

- Set up nested sampling (NS) analyses using the NS package.
- Compare the (relative) fit of two clock models, the strict clock and uncorrelated log-normal relaxed clock, to a hepatitis B virus (HBV) dataset.
- Interpret marginal likelihood estimates and Bayes' factors.

Strict and relaxed clock models

- Clock models describe variation in evolutionary rates (*e.g.*, nucleotide and morphological characters) across branches of a given tree.

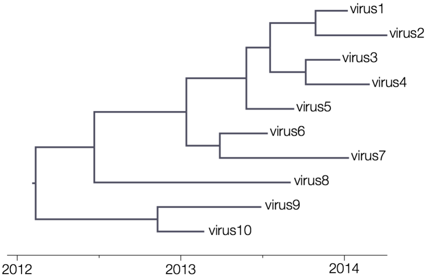

Figure: Strict clock model.

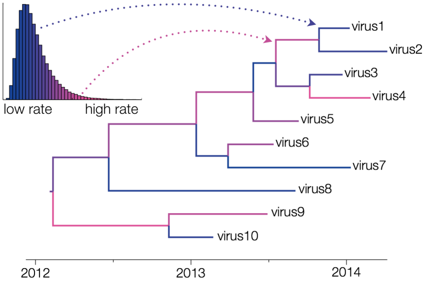

Figure: Uncorrelated LogNormal (UCLN) relaxed clock model.

Figures from https://beast.community/clocks.

Bayes' factor

We have two models, M_1 and M_2, and estimates of their (log) marginal likelihoods, Z_1 and Z_2. We can calculate the Bayes' factor (BF):

$$BF = \frac{Z_1}{Z_2}$$

$$\log(BF) = \log(Z_1) - \log(Z_2)$$

Bayes' factor

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$$BF = \frac{Z_1}{Z_2}$$

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Table: Bayes's factor (BF) range and interpretation (Kass & Raftery, 1995).

BF range	$\ln BF$ range	$\log_{10} BF$ range	Interpretation
$1 - 3$	$0 - 1.1$	$0 - 0.5$	Hardly worth mentioning
$3 - 20$	$1.1 - 3$	$0.5 - 1.3$	Positive support
$20 - 150$	$3 - 5$	$1.3 - 2.2$	Strong support
> 150	> 5	> 2.2	Overwhelming support

Questions to help you interpret ML results

1. What is marginal likelihood? What does it marginalize over?
2. Can I compare the ML estimates of models from different papers?
3. Does the ML also tell you something about the absolute fit of models to the data?
4. What about the number of parameters of the models being compared? Are parameter-rich models penalized for overfitting?